

Univariate Linear Regression from scratch using normal equation (closed form)

Residuals: A residuals is the error of single data point.

$$\text{Residual} = y_i - \hat{y}_i$$

where y_i - actual target value

$\hat{y}_i$ - value predicted by model

Sum of Residuals: $\sum y_i - \hat{y}_i$ - sum of all the residuals
positive and negative residuals try to cancel out each other.

Sum of square of residuals: To avoid cancellation, we square them
 $\sum (y_i - \hat{y}_i)^2$

Derivation:

Step 1: Define the error

$$RSS = \sum (y_i - \hat{y}_i)^2$$

where RSS - Residual Square Sum

our goal - line with smallest RSS

Step 2: We need to find slope and intercept (weights and bias of regression line)

$$\text{Let } \beta = \begin{bmatrix} b \\ w \end{bmatrix}_{2 \times 1}$$

b - bias w - weight

Step 3: matrix form of input data

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \end{bmatrix}_{n \times 2}$$

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \end{bmatrix}_{n \times 1}$$

The first column is 1 because it turns bias as another weight

Step 4:

Univariate equation (linear regression)

$$\hat{y} = w \cdot x + b$$

Step 5:

$$RSS = \sum (y_i - \hat{y})^2 \quad y_i = y$$
$$= (y - X\beta)^2$$

In matrix (vector)

$$(a-b)^2 = (a-b)^T (a-b)$$

$$(y - X\beta)^2 = (y - X\beta)^T (y - X\beta)$$

$$= (y^T - \beta^T X^T) (y - X\beta)$$

$$= y^T y - y^T X\beta - \beta^T X^T y + \beta^T X^T X\beta$$

$$y^T X\beta = [y_1, y_2, \dots, y_n]_{1 \times n} \begin{bmatrix} \quad \end{bmatrix}_{n \times 2} \begin{bmatrix} b \\ m \end{bmatrix}_{2 \times 1}$$

$$= \begin{bmatrix} \quad \end{bmatrix}_{1 \times 2} \begin{bmatrix} b \\ m \end{bmatrix}_{2 \times 1}$$

$$= \begin{bmatrix} \quad \end{bmatrix}_{1 \times 1}$$

It will result in scalar (a scalar equal to its own transpose)

$$\therefore y^T X\beta = \beta^T X^T y$$



Step 6: $(y - \hat{y})^2 = y^T y - 2\beta^T x^T y + \beta^T x^T x \beta$

Step 7: differentiate wrt β

$$\frac{\partial \text{RSS}}{\partial \beta} = 0 - 2x^T y + 2x^T x \beta$$

set $\frac{\partial \text{RSS}}{\partial \beta} = 0$

$$0 = -2x^T y + 2x^T x \beta$$

$$x^T y = x^T x \beta$$

$$\boxed{\beta = (x^T x)^{-1} x^T y} \quad \begin{array}{l} \text{(multiplied both sides} \\ \text{with } (x^T x)^{-1}) \end{array}$$